Implementation of Blind Zone and Range-Velocity Ambiguity Mitigation for Solid-state Weather Radar

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***Abstract*—This paper addresses the blind-zone problem associated with pulse compression techniques used to overcome the low peak power of solid-state transmitters, as well as inter-pulse coding techniques that take advantage of high duty cycles to mitigate range-velocity ambiguities. The implementation of these techniques on the CSU Wideband Experimental X-band (WiBEX) radar is presented.**

I. Introduction

Weather radars using low-power solid-state transmitters are an emerging technology, and promise improved reliability. These transmitters are also an essential component of active phased-array antennas being developed for the CASA project. The primary drawback of a solid-state transmitter is the low available peak power. This is because the sensitivity of a weather radar is inversely proportional to the product of the pulse width and peak power and directly proportional to the receiver bandwidth (Eq 6.55, [1]). However, solid-state transmitters have relaxed restrictions on duty cycle. Pulse compression techniques are normally used in order to achieve sensitivity comparable to a conventional electron-tube transmitter; however, this introduces a blind zone across which the radar cannot make measurements [2] [3].

At X-band, weather radars require the use of staggered-PRF waveforms in order to achieve sufficient unambiguous velocity, without compromising the maximum operational range. Waveforms developed for such applications traditionally assume the use of electron tube devices that offer high peak power, and have relatively severe duty cycle restrictions.

This paper presents the implementation of a waveform design for solid-state X-band weather radar that avoids blind-zone issues, and improves the unambiguous velocity that may be measured through these techniques.

A. Characteristics of the Radar

The waveforms presented in this paper are designed for use on the Colorado State University Wideband Experimental X-band (WiBEX) radar. This is a transportable, scanning dual-polarized X-band radar, designed to study wideband radar waveform applications for weather radar. It is currently being designed at CSU. The characteristics are presented in Table I.

TABLE I
WiBEX Radar Characteristics

Parameter	Value
Antenna	2.4 m Parabolic dual-polarized
Beamwidth	1° (3 dB)
Transmitter	Dual-channel 100 W solid-state
Frequency	9.3 - 9.5 GHz
Max. pulse width	70μs
Max. PRF	3 kHz
Transmit Waveform	Programmable multi-frequency chirp
Bandwidth	32 MHz (1 dB)
Receiver (RF)	Dual-channel single-conversion
Noise figure	2-3 dB (estimated)
Digital IF Receiver	Multi-channel pulse-compression

II. Blind Zone Mitigation

A. Nonlinear FM chirp waveform design

In order to achieve low integrated sidelobe levels (ISL), the transmitted pulse is weighted both in amplitude and in frequency taper. The frequency tapering function is designed to achieve a parametric nonlinear FM waveform. The waveform parameters are then optimized for the least ISL.

The non-linear FM chirp can be formulated as

$$u(t) = \mathrm{rect}\left(\frac{t}{T}\right)\exp\left(-j2\pi\int_0^t f(\tau)\,d\tau\right) \tag{1}$$

where T is the pulse width and $f(\tau)$ is given by

$$f(\tau) = \begin{cases} \dfrac{B}{T}\left(\dfrac{1-k_B}{1-k_T}\right), & |t| \le T(1-k_T) \quad \text{(2a)} \\ \phi(t), & |t| > T(1-k_T) \quad \text{(2b)} \end{cases}$$

where $\phi(t)$ is a nonlinear function, k_B and k_T are non-linearity parameters. Figure 1 shows the frequency taper vs. time.

B. Pulse Compression Filter Design

The received chirp signal must be filtered to obtain a compressed pulse with low ISL. This is accomplished through the use of mismatch filtering. An $\mathrm{L_p}$-norm compression filter is designed to obtain in excess of 60 dB sidelobe suppression for zero-Doppler targets [4]. Figures 2 and 3 show the results of applying the mismatch filtering to the nonlinear FM waveform from Equation 1 for various different bandwidths and Doppler velocities. The filter is seen to perform well up to a Doppler velocity of 50 m/s, which is the typical range of velocities observed by weather radars.

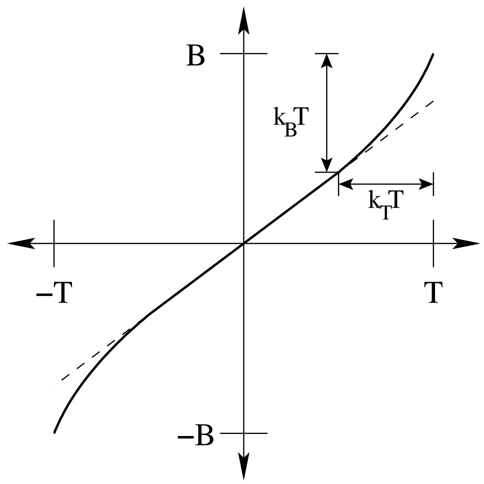

Fig. 1. Nonlinear FM characteristics

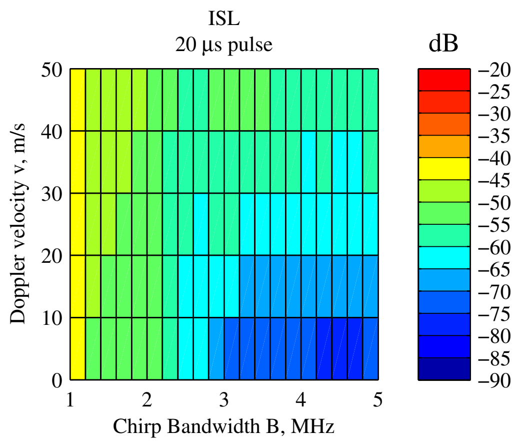

Fig. 2. Integrated sidelobe response of a 20μs chirp processed with the mismatch filter, vs. chirp bandwidth on X-axis and Doppler velocity (Y-axis)

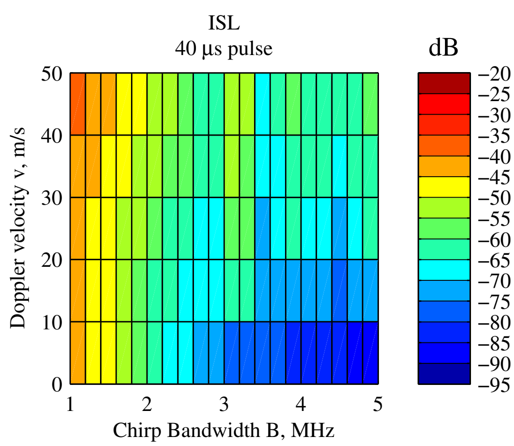

Fig. 3. Same as Figure 2, but with a 40μs chirp

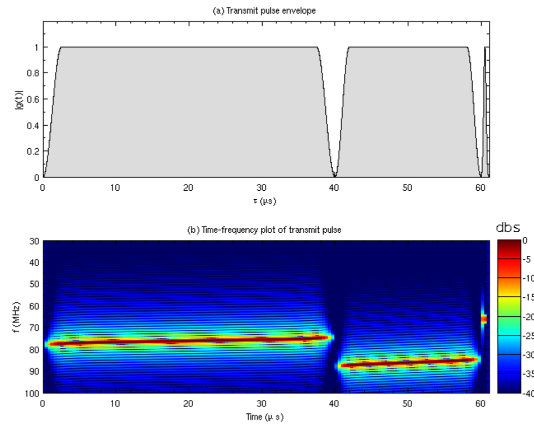

Fig. 4. Frequency-diversity transmit waveform envelope (panel a), and the time- frequency plot of the transmit waveform (panel b)

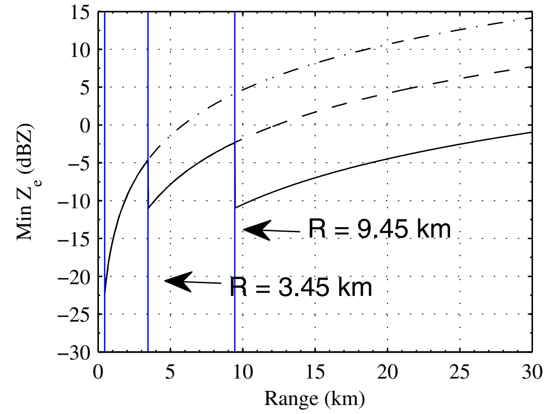

Fig. 5. Minimum detectable reflectivity for a frequency-diversity chirp waveform, with a 100 W X-band transmitter

C. Blind Zone Mitigation

With a low-peak power solid-state radar, in order to achieve sensitivity comparable to a high-power short-pulse radar, a long transmitted pulse must be used. However, since a single antenna is used on transmit and receive, the radar cannot receive signals during the transmit interval.

The blind-zone problem may be overcome through the use of a frequency-diversity transmit waveform. A set of three different pulses, each separated in time and frequency, are radiated for each pulse repetition time (PRT). Thus, dwell time is reduced compared to implementations where short and long pulses are radiated on successive PRTs. The sub-pulse widths are chosen such that the radar maintains high sensitivity for all ranges. Figure 4 shows an example of the frequency-diversity waveform. Note that in the time-frequency plot, each sub-pulse has a different center frequency.

Figure 5 shows the minimum detectable reflectivity curve for a radar using a 100 W X-band transmitter and the three-pulse frequency-diversity pulse compression waveform. The available range is divided into three segments, coverage of

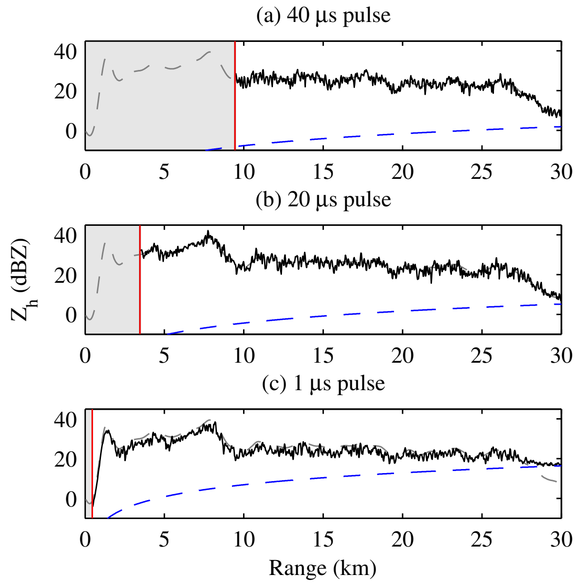

Fig. 6. Simulation results for frequency-diversity waveform. Solid black line indicates measured reflectivity, gray dashed line indicates true reflectivity. Blue dashed line is the minimum detectable reflectivity for each sub-pulse. Shaded region indicates blind zone. Data is from CSU-CHILL on June 7, 2003 during a bright-band event.

each segment is due to a different sub-pulse of the frequency-diversity waveform.

A reflectivity profile is used to simulate the performance of the frequency-diversity waveform, as shown in Figure 6. The simulation results show that for a typical reflectivity profile, the frequency-diversity waveform will retrieve the reflectivity at different ranges, including the blind-zone due to the longer pulses.

III. Range Velocity Ambiguity Mitigation

In the case of uniform pulsing techniques, the maximum range r_a and the maximum velocity v_a are related by the product $r_a \cdot v_a = c\lambda/8$, where c is the speed of light and λ is the radar wavelength. An increase in the maximum range results in a decreasing maximum velocity, and vice versa. For short wavelength X-band systems such as WiBEX, the product $c\lambda/8$ becomes small, therefore a uniform sampling scheme is not adequate to achieve the range and unambiguous velocity requirement. A non-uniform pulsing technique, namely staggered PRT with a 2:3 ratio, is used to resolve the range-velocity ambiguity problem.

The 2:3 staggered PRT waveform for simultaneous polarization transmit mode is illustrated in Figure 7. The two PRTs T_1 and T_2 are chosen as multiples of a base unit of time T_u. To mitigate the effect of second trip echoes, a random-phase coding technique is applied to the staggered PRT sequence.

$$\begin{aligned} S_r(k) &= \left[V_1(k)\, e^{j\psi_k} + V_2(k)\, e^{j\psi_{k-1}}\right] e^{-j\psi_k} && (3)\\ &= V_1(k) + V_2(k)\, e^{j(\psi_{k-1}-\psi_k)} && (4)\end{aligned}$$

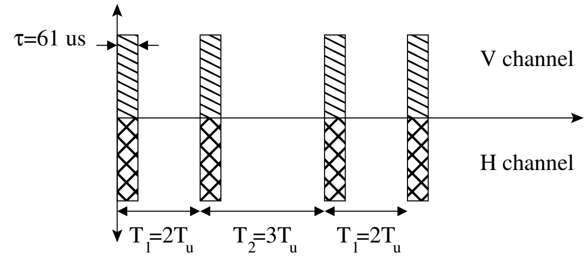

Fig. 7. Transmission scheme for 2:3 staggered PRT, in simultaneous polarization mode. Each transmitted pulse is the frequency-diversity waveform discussed in the previous section

TABLE II
Waveform Parameters

	T_u (ms)	T_1 (ms)	T_2 (ms)	v_a (m/s)	r_a (km)
Option 1	0.208	0.416	0.624	38	53
Option 2	0.138	0.276	0.414	57	32

The simulation study [7] shows that random phase coding provides good results even in the extreme case where the second trip echo is only 5 dB below the first trip echo power.

The primary disadvantage to staggered PRT waveforms is ground clutter filtering. A time-domain Gaussian Model Adaptive Processing filter (GMAP-TD) for clutter mitigation and parameter estimation is proposed for use with the WiBEX solid-state X-band radar [7]. The GMAP-TD filter works well with both uniform and 2:3 staggered PRT waveforms, and shows improvement over the spectrum-domain GMAP algorithm.

IV. Implementation

A. Transmit Waveform Generator

The frequency-diversity chirp waveform is synthesized using a custom-designed dual-channel arbitrary waveform generator. Two channels are provided to allow independent waveforms for each polarization in a dual-polarization radar.

Each sub-pulse of the frequency-diversity waveform requires approximately 10 MHz of bandwidth, including the guard band between sub-pulses. The digital baseband signal must, therefore, have a bandwidth greater than 30 MHz to accommodate 3 sub-pulses.

Figure 9 shows a block diagram of the transmit waveform generator. It is implemented using a Xilinx Spartan-3E FPGA.

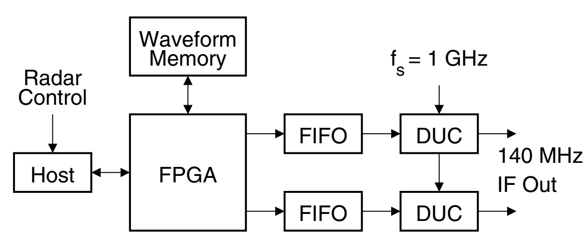

Fig. 9. Block Diagram of dual-channel arbitrary waveform generator

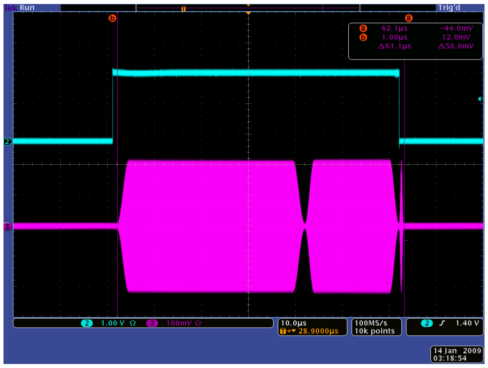

Fig. 10. Time-domain view of the waveform generator output

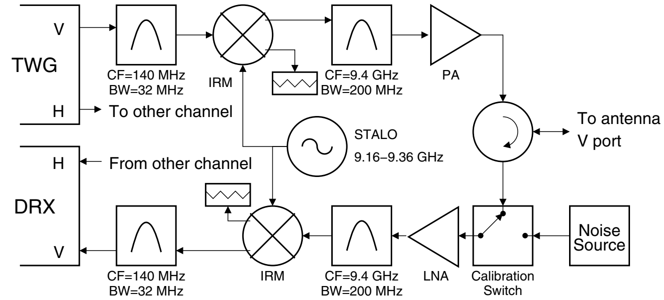

Fig. 11. RF up/down-converter block diagram. Only one polarization channel is shown. TWG is the Transmit Waveform Generator, DRX is the Digital Receiver

A host computer receives commands from the radar controller and controls the FPGA operation. Multiple chirp waveforms may be pre-computed externally at baseband, and the complex samples (I-Q pairs) are stored in the waveform memory. The FPGA reads out the waveform from memory, and stores it in a high-speed first-in-first-out (FIFO) memory during the inter-pulse interval. When triggered, the FPGA reads out samples from the FIFO and writes them to the digital up-converters (DUCs). The DUCs re-sample the data up to a 1 GSPS rate, mix with a digitally generated quadrature LO and converts the resulting samples to the analog domain.

The digital interpolation filters within the DUC limit the bandwidth of the baseband data to 80% of the sampling rate. The DAC output is filtered using a SAW reconstruction filter. The SAW filter provides very wide bandwidth, while maintaining excellent phase linearity within the passband. This is essential to prevent distortion of the wideband chirp waveform [3]. Figure 10 shows the output of the waveform generator in the time domain.

The Transmit Waveform Generator FPGA is also tasked with generating timing waveforms for the different sub-sections of the radar. Triggers for the pulsed transmitter, digital receiver and calibration switch are generated by the FPGA, using timing parameters downloaded from the host PC.

B. RF Hardware

The radar transmit RF chain is shown in Figure 11. The upconverter uses a single conversion from IF to 9.4 GHz using a single-sideband modulator. The X-band RF is then amplified using a dual-channel 100 W solid-state power amplifier. The transmitter output is then radiated through a 2.4 m dual-polarization parabolic reflector antenna. The received signals from the antennas pass through the calibration switches and the LNAs. A single-conversion receiver, using a single-sideband downconvert mixer is used to shift the signals back to the 140 MHz IF. The IF is digitized by the digital receiver ADCs for further processing.

C. Digital Receiver and Pulse Compression

All the functions related to the digital receiver are realized on a high density Xilinx Virtex-5 SXT95 FPGA available on the digital receiver board. The receiver includes four 200 MHz 16-bit analog to digital converters, the compute FPGA and a PCI interface FPGA in a PMC form-factor. The PMC is mounted on a cPCI chassis over which it transfers the I/Q output data to a single-board computer. The digital receiver implements the quadrature modulation of the sampled complex envelope of the IF signal, digital down-conversion from the IF to the baseband data rate and correlation processing.

The digital receiver is required to process all the range gates for I/Q data corresponding to each of the subpulse for both H and V channels. A traditional implementation of this 12-channel processing chain, i.e. allocating separate filters dedicated to each of the subpulses, would mandate use of excessive multipliers than the resources of the XC5VSX95T device can offer (column 2 of Table III). A multichannel design of the processing chain, wherein each filter - clocked higher than the input sample rate - shares the data for different subpulses, is chosen for implementation of the most of the functions of the digital receiver.

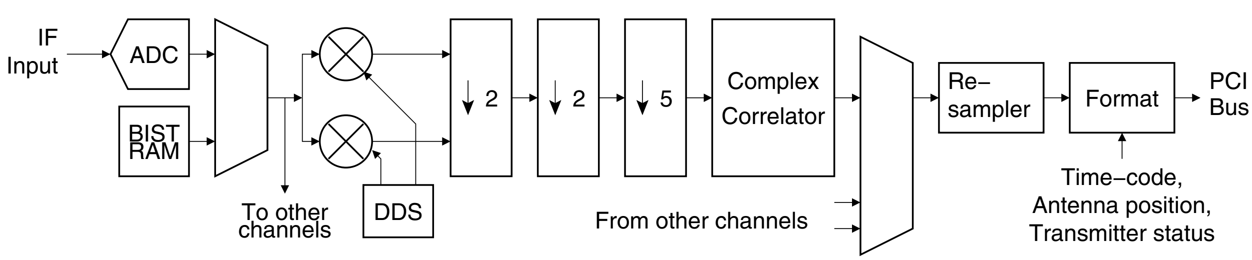

Fig. 8. Block diagram of multi-channel pulse-compression digital receiver. Processing for only one sub-pulse is shown. Two such receivers are implemented, one for each polarization channel.

TABLE III
COST OF IMPLEMENTATION OF THE FILTER CHAIN FOR ONE SUBPULSE CHANNEL. XC5VSX95T OFFERS 640 25×18 BITS DSP48E SLICES

Section	Coeffs	Mults (MATLAB)	DSP48Es (FPGA)
Halfband Filter 1	23	13	4
Halfband Filter 2	23	13	4
Decimate-by-5 Filter	255	205	13
40μs Correlation Filter	400	401	22
20μs Correlation Filter	200	201	12
Total	N/A	828 (130 %)	53 (8 %)

Figure 8 shows the major components of the digital receiver design implemented on the XC5VSX95T device. A high SFDR ($\sim$ 95 dB) multi-channel direct digital synthesizer (DDS) is employed for the quadrature modulation of the sampled IF data. A cascade of the half-band filters and a low pass FIR filter decimates the I/Q data so obtained. Polyphase implementation of these filters further reduces the usage of DSP48E slices. The matched filtering of the I/Q data is performed by single-rate FIR filters clocked higher than the input sample rate. This entire design consumes close to 400 DSP48E slices on the FPGA. Column 3 in Table III shows the total number of DSP48Es used for the implementation of the filter chain for one subpulse channel.

This implementation also offers high programmability as several design parameters, e.g. the coefficients for the filters, can be modified on-the-fly depending on the waveform used. The digital receiver also provides for a reliable self-test mechanism through a programmable Built-In Self Test (BIST) and online digital health report (by checking status of output at a simulated target placed at an unusable range gate). Some other auxiliary functions performed by the digital receiver include extracting time information from an IRIG-B encoded pulse and angle information from the encoded antenna positioner signal.

V. CONCLUSION

A frequency-diversity chirp waveform for use with solid-state X-band weather radar has been developed. While it is expected theoretically to be a viable solution, the practical implementation issues of the frequency-diversity waveform have not been addressed before in the weather radar context. The key objective is to implement this technique in hardware and establish practical limits.

The waveform has been successfully implemented in test versions of the hardware, and we are awaiting development of the final versions for testing and validation against existing reference radars.

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